

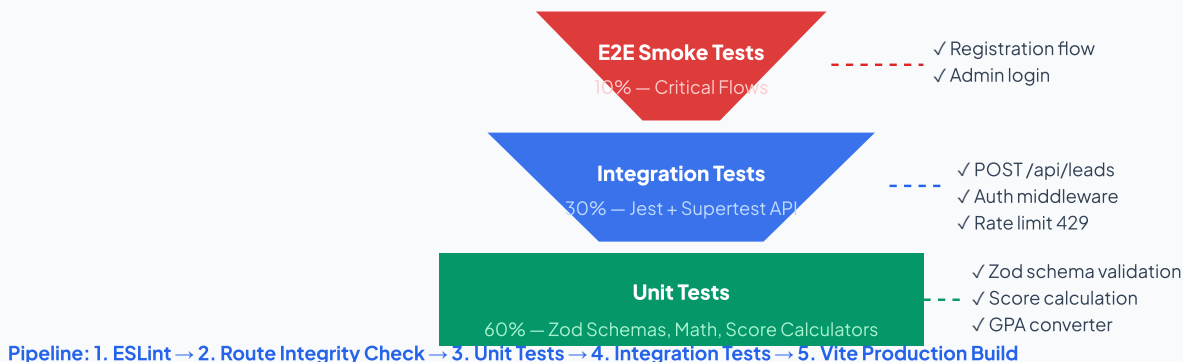
Testing & Quality Assurance

Testing Pyramid architecture, Jest + Supertest API integration suites, module integrity checkers, deterministic mock strategy, and build-gate verification.

🔧 Runner: Jest + Supertest 🚦 Gate: Module Integrity Checker ✅ 32 Routes Verified

1. Testing Pyramid & Pipeline Architecture

▲ UNICOACH TESTING PYRAMID — 3-TIER QA STRATEGY



🔗 SUPERTEST IN-MEMORY API TESTING — NO PHYSICAL NETWORK PORTS



```
// tests/lead.test.js - Supertest Integration Suite
jest.mock('../src/services/notification.service', () => ({
  dispatchLeadConfirmation: jest.fn().mockResolvedValue(true),
}));
describe('POST /api/leads', () => {
  it('returns 201 for valid lead payload', async () => {
    const res = await request(app).post('/api/leads').send({
      name: 'John',
      email: 'j@test.com',
      phone: '+919876543210'
    });
    expect(res.status).toBe(201);
    expect(res.body.success).toBe(true);
  });
  it('returns 400 for invalid phone format', async () => {
    const res = await request(app).post('/api/leads').send({
      name: 'John',
      email: 'j@test.com',
      phone: 'bad-phone'
    });
    expect(res.status).toBe(400);
  });
});
```

2. Technology Decision Framework

1. Jest + Supertest vs Mocha + Chai vs Postman

TEST FRAMEWORK

WHAT IS IT?

Jest is a zero-configuration parallel test runner with built-in assertions and mocking. Supertest wraps an Express app for in-memory HTTP testing without binding to a network port.

HOW DOES IT WORK & PROBLEM SOLVED?

Binding to a real TCP port during tests causes random port collisions in CI environments and adds 100–500ms latency per test suite. Supertest passes synthetic requests directly into the Express application handler in-memory, completing in milliseconds.

WHAT ALTERNATIVES EXISTED?

Mocha + Chai: Requires separate test runner, assertion library, spy library, and coverage tool. **Manual Postman:** Not automatable in CI pipelines.

WHY DID WE CHOOSE THIS APPROACH?

Jest ships with everything: parallel execution, mocking, coverage, watch mode. Supertest integrates directly with Express. Zero configuration overhead.

💡 Senior Interview Defense: "How do you ensure code quality before releases?"

"UniCoach uses a 5-stage quality pipeline. ESLint runs first for static analysis. Then an automated module integrity checker verifies all 32 Express routes map to existing controller functions — catching the classic 'callback got undefined' boot crash before it reaches production. Jest unit tests cover Zod schemas and utility functions. Supertest integration suites validate every API endpoint in-memory with Twilio and Groq mocked out. Finally, a mandatory production build gate runs Vite on both frontend and admin apps — if any syntax error or broken import exists, the build fails and the git push is blocked."